

Exercise 03

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Problem 1a.

Solution. Note

$$\mathbb{P}_x(H_y < \infty, H_y > n \mid X_n = z) = \mathbb{P}_x(H_y < \infty \mid X_n = z)\mathbb{P}_x(H_y > n \mid H_y < \infty, X_n = z)$$

and

$$\mathbb{P}_x(H_y < \infty, H_y > n \mid X_n = z) = \mathbb{P}_x(H_y > n \mid X_n = z)\mathbb{P}_x(H_y < \infty \mid H_y > n, X_n = z)$$

1. $\mathbb{P}_z(H_y > n)\mathbb{P}_x(H_y < \infty \mid X_n = z)$: $\mathbb{P}_z(H_y > n) \neq \mathbb{P}_x(H_y > n \mid X_n = z)$. Thus this product isn't equal to the given.
2. $\mathbb{P}_x(H_y < \infty)\mathbb{P}_z(H_y > n \mid X_n = x)$: Not equal
3. $\mathbb{P}_z(H_y < \infty)\mathbb{P}_x(H_y > n \mid X_n = z)$: Note $\mathbb{P}_x(H_y < \infty \mid H_y > n, X_n = z) = \mathbb{P}_z(H_y < \infty)$ by the strong markov property. Thus this product is equal.
4. $\mathbb{P}_x(H_y > n)\mathbb{P}_z(H_y < \infty \mid H_n = x)$: Not equal
5. $\mathbb{P}_x(H_y < \infty, H_y > 2n \mid X_{2n} = z)$: Not equal

■

Problem 1b.

Solution. 1. Doesn't hold

2. Holds
3. Holds
4. Doesn't Hold
5. Holds

■

Problem 1c.

Solution. 1. Holds

2. Holds

3. Holds
4. Doesn't Hold
5. Holds

■

Problem 3.2a.

Proof.

$$\begin{aligned}
E_\mu[f((X_{k+n})_{n \geq 0}) \mid \mathcal{F}_k] &= E_\mu[f((X_{k+n})_{n \geq 0}) \mid (X_1, \dots, X_k)] \\
&= E_\mu[f((X_{k+n})_{n \geq 0}) \mid X_k] \\
&= E_{X_k}[f((X_{k+n})_{n \geq 0})] \\
&= g(X_k)
\end{aligned}$$

□

Problem 3.2b.

Proof.

$$\begin{aligned}
E_\mu[f((X_{T+n})_{n \geq 0}) \mid \mathcal{F}_T] &= \sum_{m=1}^{\infty} 1_{T=m} E_\mu[f((X_{T+n})_{n \geq 0}) \mid \mathcal{F}_T] \\
&= \sum_{m=1}^{\infty} 1_{T=m} E_\mu[f((X_{T+n})_{n \geq 0}) \mid \mathcal{F}_m] \\
&= \sum_{m=1}^{\infty} 1_{T=m} g(X_m) \\
&= g(X_T)
\end{aligned}$$

□

Problem 3.3a.

Proof. Let $h_N(x) : [0, \dots, N] \rightarrow [0, 1]$ where $h_N(x) = P_x(\tilde{H}_N < \tilde{H}_0)$ for $N \in \mathbb{N} \setminus \{0\}$.

1. $h(0)$: Note $\mathbb{P}_0(\tilde{H}_0 = 0) = 1$. Thus $h(0) = \mathbb{P}_0(\tilde{H}_N < \tilde{H}_0) = \mathbb{P}_0(\tilde{H}_N < 0) = 0$
2. $h(N)$: Note $\mathbb{P}_N(\tilde{H}_N = 0) = 1$. Thus $h(N) = \mathbb{P}_N(\tilde{H}_N < \tilde{H}_0) = \mathbb{P}_N(0 < \tilde{H}_0) = 1$
3. For $x \in [1, \dots, N-1]$:

$$\begin{aligned}
h(x) &= \mathbb{P}_x(\tilde{H}_N < \tilde{H}_0) \\
&= \mathbb{P}_x(X_1 = x-1) \mathbb{P}_x(\tilde{H}_N < \tilde{H}_0 \mid X_1 = x-1) + \mathbb{P}_x(X_1 = x+1) \mathbb{P}_x(\tilde{H}_N < \tilde{H}_0 \mid X_1 = x+1) \\
&= \mathbb{P}_x(X_1 = x-1) \mathbb{P}_{x-1}(\tilde{H}_N < \tilde{H}_0) + \mathbb{P}_x(X_1 = x+1) \mathbb{P}_{x+1}(\tilde{H}_N < \tilde{H}_0) \\
&= \frac{1}{2} h(x-1) + \frac{1}{2} h(x+1)
\end{aligned}$$

□

Problem 3.3b.

Proof. Thus $-2h(x) + h(x-1) + h(x+1) = 0$. The characteristic equation is $\lambda^2 - 2\lambda + 1 = 0 \implies (\lambda - 1)^2 = 0$. Thus $h(x) = A + Bx$. $h(0) = A = 0$ and $h(N) = BN = 1 \implies B = \frac{1}{N}$. Thus $h(x) = \frac{x}{N}$. □

Problem 3.4.

Proof. Let $y \in S$ where $n(y) = N$. For $x \in S$:

$$\begin{aligned} \mathbb{P}_x(\tau_C \leq N) &\geq \mathbb{P}_x(\tau_C \leq n(x)) \\ &\geq \mathbb{P}_y(\tau_C \leq N) \\ &\geq \epsilon \end{aligned}$$

Thus $\mathbb{P}_x(\tau_C > N) = 1 - \mathbb{P}_x(\tau_C \leq N) \leq \mathbb{P}_y(\tau_C > N) = 1 - \mathbb{P}_y(\tau_C \leq N) \leq 1 - \epsilon$.
Suppose for some $k \in \mathbb{N}$, $\mathbb{P}_x(\tau_C > kN) = (1 - \epsilon)^k$

$$\begin{aligned} \mathbb{P}_x(\tau_C > (k+1)N) &= \mathbb{P}_x(\tau_C > kN) \mathbb{P}_x(\tau_C > (k+1)N \mid \tau_C > kN) \\ &\geq (1 - \epsilon)^k \mathbb{P}_y(\tau_C > N) \\ &\geq (1 - \epsilon)^{k+1} \end{aligned}$$

□

Problem 3.5a.

Proof. $\forall n \in 2\mathbb{N}$ and $\forall a \in \mathbb{N} \ni a \bmod 2 = 1$. Let $A = \max(X_m : 0 \leq m \leq n)$ and $B = \max(X_m : 0 \leq m < n)$

$$\begin{aligned} \mathbb{P}_0[A \geq a] &= \mathbb{P}_0(A \geq a \mid X_n > a) \mathbb{P}(X_n > a) + \mathbb{P}_0(A \geq a \mid X_n \leq a) \mathbb{P}_0(X_n \leq a) \\ &= \mathbb{P}(X_n > a) + \mathbb{P}_0(A \geq a \mid X_n \leq a) \mathbb{P}_0(X_n \leq a) \\ &= \mathbb{P}(X_n > a) + \mathbb{P}_0(A \geq a, X_n \leq a) \end{aligned}$$

Note the following, that since a is odd, $a = 2k+1$ for some $k \in \mathbb{N}$. Let x be the number of $+1$ moves and y be the number of -1 moves. Thus $x + y = n$ and $x - y = a$. Thus $2x = n + a = n + 2k + 1 \implies x = \frac{n}{2} + k + \frac{1}{2}$. Note n is even so $\frac{n}{2} \in \mathbb{Z}$ so $x \notin \mathbb{N}$. Thus $X_n \neq a$.

Thus $\mathbb{P}_0(A \geq a, X_n < a) = \mathbb{P}_0(A \geq a, X_n \leq a)$. Note if $A \geq a$ and $X_n < a$ imply that $\exists m \in [0, n] \cap \mathbb{N} \ni X_m = a$. Thus $\mathbb{P}_0(A \geq a, X_n < a) = \mathbb{P}_0(H_a \leq n, X_n < a)$. Thus,

$$\begin{aligned} \mathbb{P}_0[A \geq a] &= \mathbb{P}_0(X_n > a) + \mathbb{P}_0(A \geq a, X_n \leq a) \\ &= \mathbb{P}_0(X_n > a) + \mathbb{P}_0(H_a \leq n, X_n < a) \end{aligned}$$

□

Problem 3.5b.

Proof.

$$\begin{aligned}
\mathbb{P}_0(H_a \leq n, X_n < a) &= \sum_{m \in \mathbb{N}} \mathbb{P}_0(H_a = m) \mathbb{P}(H_a \leq n, X_n < a \mid H_a = m) \\
&= \sum_{0 \leq m \leq n} \mathbb{P}_0(H_a = m) \mathbb{P}(H_a \leq n, X_n < a \mid H_a = m) \\
&= \sum_{0 \leq m \leq n} \mathbb{P}_0(H_a = m) \mathbb{P}(X_n < a \mid H_a = m)
\end{aligned}$$

Let $Y_k = X_{k+m} - a$.

Thus $\mathbb{P}_0(X_n < a \mid H_a = m) = \mathbb{P}_0(X_n < a \mid H_a = m, X_0 = 0) = \mathbb{P}_0(X_n < a \mid X_m = a) = \mathbb{P}(Y_{n-m} < 0 \mid Y_0 = 0) = \mathbb{P}_0(Y_{n-m} < 0) = \mathbb{P}_0(Y_{n-m} > 0)$ where the second equality is 1 step markov property and last equality is because of the reflexive property of SRW. Then $\mathbb{P}_0(Y_{n-m} > 0) = \mathbb{P}(Y_{n-m} > 0 \mid Y_0 = 0) = \mathbb{P}(X_n > a \mid X_m = a) = \mathbb{P}(X_n > a \mid H_a = m) = \mathbb{P}_0(X_n > a \mid H_a = m)$.

Thus

$$\begin{aligned}
\mathbb{P}_0(H_a \leq n, X_n < a) &= \sum_{0 \leq m \leq n} \mathbb{P}_0(H_a = m) \mathbb{P}(X_n < a \mid H_a = m) \\
&= \sum_{0 \leq m \leq n} \mathbb{P}_0(X_n > a \mid H_a = m) \mathbb{P}_0(H_a = m) \\
&= \mathbb{P}_0(X_n > a)
\end{aligned}$$

Thus

$$\begin{aligned}
\mathbb{P}_0[A \geq a] &= \mathbb{P}_0(X_n > a) + \mathbb{P}_0(H_a \leq n, X_n < a) \\
&= \mathbb{P}_0(X_n > a) + \mathbb{P}_0(X_n > a) \\
&= \mathbb{P}_0(X_n < -a) + \mathbb{P}_0(X_n > a) \\
&= \mathbb{P}_0(|X_n| > a) \\
&= \mathbb{P}_0(|X_n| \geq a)
\end{aligned}$$

The third equality comes from reflection property of simple random walk and last equality comes from the fact that $\mathbb{P}_0(|X_n| = a) = 0$ since n is even and a is odd. $\square$

Problem 3.6a.

Proof. Let $k_i = E_i(H_9)$. Note the computation of k_8, k_9, k_6, k_3, k_2 is redundant as they are the heads of snakes, bases of ladders, or finish so a user doesn't stay there for a turn. We get the following relations after:

$$\begin{aligned}
k_1 &= \frac{1}{2}k_7 + \frac{1}{2}k_5 + 1 \\
k_4 &= \frac{1}{2}k_5 + \frac{1}{2}k_1 + 1 \\
k_5 &= \frac{1}{2}k_1 + \frac{1}{2}k_7 + 1 \\
k_7 &= \frac{1}{2}k_4 + 1
\end{aligned}$$

Solving the relations we see $k_1 = 7$, $k_4 = 8$, $k_5 = 8$, $k_7 = 5$. Thus on average it takes 7 turns to finish the game. $\square$

Problem 3.6b.

Proof. The question is effectively asking to find for $i = 5$, $\mathbb{P}_i(H_1 > \infty)$. Let $f_i = \mathbb{P}_i(H_1 > \infty)$. Note $f_9 = 1$, $f_1 = 0$. Note the computation of f_8 , f_6 , f_3 , k_2 is redundant as they are the heads of snakes, bases of ladders, or finish so a user doesn't stay there for a turn.

$$\begin{aligned}
f_1 &= 0 \\
f_4 &= \frac{1}{2}f_5 \\
f_5 &= \frac{1}{2}f_7 \\
f_7 &= \frac{1}{2}f_4 + \frac{1}{2}
\end{aligned}$$

Thus $f_5 = \frac{2}{7}$. Thus $\mathbb{P}_5(H_1 > \infty) = \frac{2}{7}$. $\square$